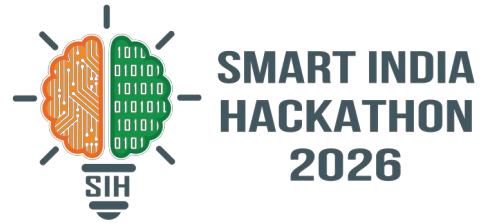


SMART INDIA HACKATHON 2026



KrishiSetu

Problem Statement ID – SIH26033

Problem Statement Title – **Farmer-first agricultural trading platform with pooled delivery**

Theme – **Agriculture, FoodTech & Rural Development**

PS Category – **Software**

Team ID – **SIH26-SW059**

Team Name – **Logic_Lords — Ramrao Adik Institute of Technology**



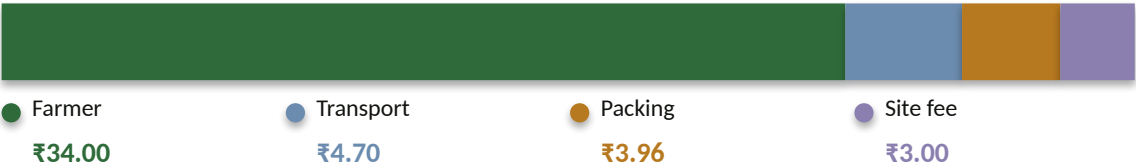
The farmer names the price.
Everything after it is shared transport.

Working prototype · krishisetu-weld.vercel.app



The farmer's rate is an input to the price, never the leftover

ONE KILOGRAM OF TOMATO, BUILT UPWARD



₹45.66 buyer pays vs ₹55.00 quick-commerce

ONE 5 KG ORDER • TWO NAMED FARMS

	Sanjay Patil Khalapur, Raigad	3.0 kg	₹102.00
	Meena Bhoir Panvel Rural, Raigad	2.0 kg	₹68.00

WHAT IS ACTUALLY DIFFERENT



Net rate
nothing deducted, ever



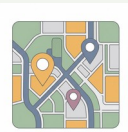
Pooled by place
one neighbourhood, one vehicle



Sold before loaded
no stock written off



Fresher, cheaper
shorter chain, lower price



Demand by place
Nerul 246 • Kharghar 207



Surplus re-placed
151 kg split across three

HOW THE BUYER RECEIVES IT



Cluster pickup
₹10



Last 300 m
₹20

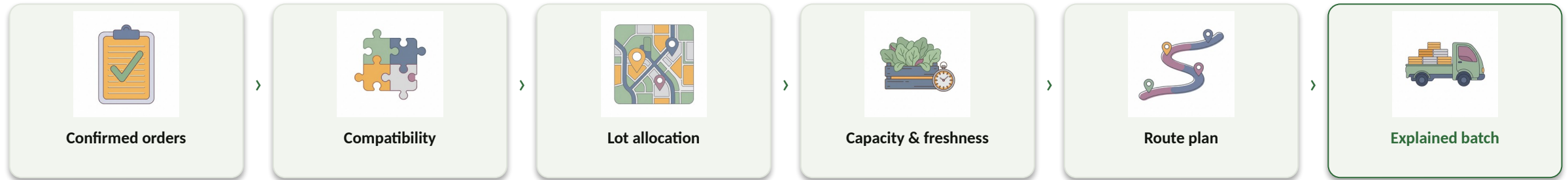


Express
not offered

Free above a ₹500 basket on the last 300 m. ₹55/kg is an assumed quick-commerce reference, not a live scrape.

D-COA — Decision & Coordination Optimisation Algorithm

Our own engine's working name, not a claim of a new published algorithm.



EIGHT CONSIDERATIONS, EVERY RUN

● Product compatibility	hard	● Pickup & delivery geography	hard
● Quantity vs vehicle capacity	hard	● Delivery time windows	hard
● Perishability & freshness	hard	● Estimated transport cost	ranking
● Order priority	ranking	● Cancellations & new orders	replan



Next.js 16 · PostgreSQL · Prisma
Vercel · live mandi prices, daily

RULES THE CODE ENFORCES

- Under 70% fill, the run is held
- Atomic reservations: no kilogram twice
- A forecast shows only if it beats the baseline

Routing returns a feasible nearest-neighbour plan measured against a fixed-order baseline. We claim no global optimum.

FEASIBILITY AND VIABILITY

Return loads, the fill threshold, and what can go wrong

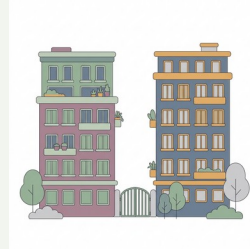
RETURN LOADS — THE EMPTY HALF OF EVERY TRIP



loaded 747.7 kg ›

‹ returns empty today

The empty half is already priced into what he charges us.



123

orders pooled

747.7 kg

on a 750 kg tempo

5

stops: 4 farms, 1 drop

₹5.37

transport per kg

One real run, planned by the live system

Return-load matching is designed and specified. It sits deliberately after the core proof and is not claimed as shipped.

RISK, AND WHAT IS BUILT AGAINST IT



Too few orders

Held below 70% fill, rolls to the next window



Spoilage in transit

Freshness is a hard constraint, per commodity



Quality disputes

Graded, handover recorded, deduction by agreement



Farmer cannot use an app

Marathi, Hindi, English · one decision card

IMPACT AND BENEFITS

More for the farmer, less for the buyer, and still self-funding

FARMER'S SHARE OF WHAT THE BUYER PAYS



The fragmented-chain figure is a widely reported estimate. The two KrishiSetu bars are computed from orders in the running system.



WHERE OUR OWN MONEY COMES FROM

Disclosed buyer-side fee, per order	₹19.89
Gateway, messaging, support	– ₹5.60
Contribution per order	₹14.29
Assumed monthly fixed cost	₹60,000
Orders a month to break even	4,199

No listing fee. No commission. To hold a price, our fee is cut to zero first — never the farmer's amount.



Farmer
₹34.00/kg, paid in full



Buyer
₹45.66/kg vs ₹55 reference



Transporter
₹4,016 a run, agreed first



Platform
₹14.29, one disclosed fee

Figures are computed over synthetic demonstration orders: they size the model rather than measure a business. No column here is farming profit.

RESEARCH AND REFERENCES

Sources, and what they do and do not prove



Government of India open data portal

data.gov.in — daily mandi prices

LIVE IN APP



eNAM — trading and FPO pages

enam.gov.in/web/trading-details

INCUMBENT



Ninjacart and Ninja Mandi

ninjacart.com/ninja-mandi

INCUMBENT



agribazaar

agribazaar.com/quick-links/steps

INCUMBENT



Samunnati

samunnati.com/innovations

INCUMBENT



Google OR-Tools, VRP with time windows

developers.google.com/optimization/routing/vrptw

METHOD



PIB release on agricultural marketing

pib.gov.in — PRID 2222802

POLICY



SIH 2026 idea submission format

sih.gov.in/letters/2026

FORMAT

What this does not prove: demand history and the fitted price series are synthetic and labelled so on screen. An unmentioned capability on a competitor's page is not evidence they lack it, and we claim no measured superiority without a comparable pilot.